INCH-POUND

A-A-54894
23 July 1993

COMMERCIAL ITEM DESCRIPTION

DEFIBRILLATOR, MONITOR-RECORDER

(Battery Operated, with Charger)

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-D-42054A, which is cancelled.

This Commercial Item Description covers a battery powered defibrillator/monitor-recorder with battery charger suitable for use on patients experiencing cardiac crisis in military combat zone environments.

Salient characteristics:

The unit shall be an integral or modular portable ECG monitor/recorder and defibrillator which functions from self-contained batteries. Units which are modular in design and can be separated shall consist of only two separate components: a defibrillator and a monitor-recorder. The device shall be capable of synchronous cardioversion. It shall possess a monitor for examining the electrical activity of the heart and a means to produce a permanent record of the heart activity in the three standard ECG lead format. Each component shall be independently powered and shall securely interconnect into a physically contiguous unit without the use of tools, without the need for auxiliary cables, and in such a manner as not to cause excessive stress on the electrical connectors.

The unit shall include batteries, defibrillator paddles, synchronized cardioversion capability, and accessories as specified. Provision shall be made for out of the way paddle storage. The unit shall also include a battery charger which shall be separate from the defibrillator and monitor-recorder.

AMSC N/A FSC 6515

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

The unit, whether modular or not, shall comply with applicable sections and requirements of the following standards except as otherwise defined in this specification:

* American National Standards Institute\Association for the Advancement of Medical Instrumentation DF2, ANSI\AAMI DF2 - American National Standard for Cardiac Defibrillator Devices, with the exception of paragraph 3.2.1.2, Energy Range

* Underwriters' Laboratories Standard 544, UL544 - Standard for Safety of Electric Medical and Dental Equipment

* National Fire Protection Association 99, NFPA 99 - Standard for Health Care Facilities

* American Heart Association, AHA Publication titled - Recommendations for Standardization of Leads and of Specifications for Instruments in Electrocardiography and Vectorcardiography

* Association for the Advancement of Medical Instrumentation ECG 5/83, AAMI ECG 5/83 - Standard for ECG Connectors

* International Electrotechnical Commission-Safety of Medical Electrical Equipment, (IEC-601-2-4, Sealed Case)

The unit shall be capable of full operation while still providing environmental protection and shall comply with environmental conditions as follows:

Atmospheric Pressure:	400 - 700 mm Hg
Relative Humidity:	95 percent (non-condensing)
Temperature:	
Operational:	$32^{o}F$ - $114^{o}F$
Storage:	$-23^{o}F$ - $149^{o}F$

All units shall be easily cleaned with routine hospital cleansing solutions, adequately protected from moisture, and be designed to withstand normal use, such as shock and impact, frequently encountered in circumstances outside the normal hospital environment. The unit shall be capable of being fully operated while still providing environmental protection from moisture, dust, and mud.

The unit shall contain an integral 50 ohm test load which may be incorporated in either the defibrillator module or charger module.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Each module shall incorporate a visual and\or audio low battery warning.

The unit shall be capable of withstanding the drip proof test specified without damage and shall operate properly in all modes following the test. All controls shall be immediately accessible (no cover) during all steps of the test.

The total weight of the defibrillator, monitor-recorder including the batteries, paddles, etc., shall not exceed 30 pounds.

The monitor-recorder shall have the following characteristics:

* It shall be capable of providing an ECG through the defibrillator paddles or via a 3-lead patient cable having a six pin cable connector.

* The patient cable shall be shielded and have an approximate minimum length of 10 feet. Individual lead length shall be approximately 3 feet. Total cable length with patient leads shall not exceed 13 feet.

The cardioscope display shall have the following characteristics:

* It shall have a non-fade type display measuring no less than 1.4 by 2 inches.

* It shall have either a freeze control or a delayed recording capability for detailed analysis of ECG arrhythmia.

* It shall display a signal in the monitor mode with a measurable low frequency cutoff point (-3 dB down) at 1.0 Hz, and a high frequency cutoff point (-3 dB down) of 30 Hz.

* It shall have a sweep speed of one inch/second.

* It shall have an easily adjustable audible systole indicator.

* It shall have a lead selector switch enabling the operator to select lead I, II, and III without switching patient electrode sites.

* ECG gain shall be continuously adjustable or discrete step mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

The ECG amplifier shall have the following characteristics:

* It shall be protected from defibrillator counter shock up to a level of 360 joules.

* It shall have, as a minimum, common mode rejection ratio of 80 dB with respect to chassis ground with 65 kilohms ohms imbalance at 60 Hz.

* It shall have an electrically isolated patient input.

* It shall have a 1 mV reference signal for ECG calibration.

The recorder shall have the following characteristics:

* The frequency response shall conform at a minimum to the specifications given to the cardioscope display.

* It shall have a paper speed of one inch/second. Paper speed shall be accurate to within plus or minus 5 percent.

* The ECG paper size shall have a minimum grid width of 1.57 inch, and be at least 50 feet long.

* A low paper indication shall be provided, either by the instrument or in the form of a mark on the paper.

* The recorder shall be an integral part of the monitor section of the defibrillator, monitor-recorder and not a free standing instrument requiring electrical cabling interconnection.

The defibrillator shall have the following characteristics:

* The pulse shape shall be a clinically proven one such as Lown, Edmark, or Trapezoidal.

* The maximum selectable delivered energy of the defibrillator shall lie in the range of 320 to 400 joules, inclusive. It shall provide a selection of at least six energy outputs within the range of 5 joules to maximum energy when delivered into a 50 ohm resistive load. At least three outputs shall be less than 110 joules.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

* The delivered energy pulse shall be within 4 joules or plus or minus 15 percent of the selected energy on the energy level indicator, whichever is greater, when discharged into a 50 ohm resistive load.

* The defibrillator shall be capable of delivering ten pulses at maximum energy into a 50 ohm resistive load within five minutes, at 32°F.

* The paddles shall conform to AHA dimensions and be capable of ECG pickup through the paddle surface. Both paddles shall be provided with discharge buttons and both shall be activated simultaneously to discharge the unit. One paddle shall be provided with a remote charge button.

* It shall have a maximum charge time to maximum energy setting of 12 seconds or less at 82.5°F.

In addition to the required accessories above, the following shall be provided, when specified by the contracting officer:

* External pediatric paddles - one pair required. The paddles shall provide at least a 15 square centimeter contact area per electrode.

The unit shall have a synchronizer with visual indication on the cardioscope to identify the synchronizer trigger point when the cardioversion mode of the defibrillator is activated. Unit must switch back to defibrillation mode after each cardioversion attempt. The time between the leading edge of a square pulse and of the defibrillator's syncronized discharge pulse shall not exceed 30 ms.

It shall provide a visible means of determining selected energy by a calibrated knob setting, digital display, or analog meter.

It shall provide a visual ready indicator and audible tone to indicate when the defibrillator is fully charged and ready for discharge.

The ECG defibrillator and monitor-recorder shall operate from self-contained batteries. Systems in which the defibrillator and monitor-recorder can be used separately shall be independently powered, each containing an internal battery source.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

The defibrillator, monitor-recorder charging system shall operate over the full tolerance range of one or more of the following line voltage and frequency combinations as specified by the customer requisition:

120 VAC, plus or minus 10 percent, 50 Hz, plus or minus 2 Hz
120 VAC, plus or minus 10 percent, 60 Hz, plus or minus 2 Hz
240 VAC, plus or minus 10 percent, 50 Hz, plus or minus 2 Hz

Dual voltage and hertz on all units is desirable but not mandatory.

Changeover devices, switches, circuits, or transformers, if used for the purpose of voltage or frequency conversion, shall be clearly marked and identified in the appropriate documentation which accompanies the unit. Mechanical or electrical adjustment to effect the conversion shall not require the use of specialized tools or skills.

The monitor-recorder shall operate from fully charged batteries for a minimum of 2.25 consecutive hours of cardiac monitoring, or a minimum of 45 consecutive minutes of recorder operation, or any linear combination of both.

The defibrillator shall operate from fully charged batteries for a minimum of 30 shocks at maximum energy at 82.5°F. The charge time to maximum energy output shall not exceed 12 seconds at 82.5°F. The unit shall deliver ten shocks at maximum energy into a 50 ohm resistive load at 32°F within five minutes.

The charge time for fully depleted batteries shall not exceed 16 hours. The unit shall have a charge level indicator, a charging lamp/indicator, and a low voltage indicator for each battery group.

The batteries shall be quick-change type.

All plugs shall be hospital grade.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Accessories shall include the following:

Two patient cables
One spare stylus (with replacement tool if required)
One ECG ruler
One box pre-gelled electrodes
Two spare battery packs of each type used
Three packages recording paper w/NSN
One tube electrode jelly
One accessory bag

The unit shall be of a rugged construction capable of withstanding hazards of normal use in a hospital setting as well as those found in field conditions.

The unit shall have a code summary critical event memory that can document up to 20 of the following:

* 8.5 second defibrillation ECG segments (three second pre-shock, 5.5 second post shock).

* Six seconds of recorded ECG data.

* Summary of elapsed time, number of shocks delivered, date, and real time.

The contractor shall furnish manuals, and/or brochures containing complete operation, installation and service/maintenance instructions (including pictures or illustrations, as necessary) with complete schematics and wiring diagrams. All manuals, handbooks, and/or brochures shall be written in the English language and all schematics and wiring diagrams shall use American electrical and electronics symbols. The manuals shall include electrical data and connection diagrams for all applicable utilities. The instructions shall also contain a complete list of all replaceable parts showing part number, name, and quantity required. All service notes, service memos, etc., pertaining to the equipment and issued subsequent to the printing of the original instructions shall also be included. These instructions shall be, as a minimum, that furnished to service engineers (or servicemen) that normally install and service the equipment for the company or distributor. When the system being procured includes, or will operate ancillary equipment, the service data shall include complete instructions and drawings which show interfacing of all system components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Two complete sets of the above information shall be packaged with each unit of the required equipment. Where multiple quantities of the same unit are to be delivered to the same ultimate consignee, then only a total of two service manuals need be supplied to each consignee. However, two operation manuals still need to be supplied with each unit.

Workmanship. The defibrillator, monitor-recorder shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One defibrillator, monitor-recorder, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of S-1. The unit of product for examination purposes shall be one defibrillator, monitor-recorder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 1.0 (percent defective) and an inspection level of S-2. The unit of product for test purposes shall be one defibrillator, monitor-recorder.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall include, but shall not be limited to, the following:

Pull test. Gradually apply a 35 pound pull in a direction that would be the most likely to pull the cord from the enclosure and maintain for one minute. The strain relief is not acceptable if there is such movement of the cord as to indicate that stress is being transmitted to terminals, splices or interior wiring.

Measurement test. Measurements shall be made, using an appropriate measuring instrument, to verify isolated patient connected circuitry is spaced at a minimum of 1/8 inch from non-isolated circuitry.

Leakage current test. Leakage current measurements shall be made with power on and off, and defibrillator in both charged and uncharged states.

Chassis leakage current test. Measured between any exposed conductive surface and ground shall not exceed 100 micro amperes.

Paddle leakage current test. Measured between an external paddle electrode and ground shall not exceed 100 micro amperes. Leakage current measured between an internal paddle electrode and ground shall not exceed 50 micro amperes.

Lead (source) leakage current test. Measured between each lead and ground, all leads and ground, and any two leads shall not exceed 10 micro amperes.

Lead (sink) current test. Measured for each lead at a potential of 120Vac to ground shall not exceed 20 micro amperes.

(All leakage current measurements shall be performed with an AAMI standard test load).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Paddle handle test. (When wrapped with aluminum foil) leakage current, to the paddle electrode, shall not exceed 250 micro amperes when a voltage 1.5 times the maximum energy storage capacitor voltage (1.06 times if 60 Hz) is impressed between handle and electrode.

Paper speeds test. Measure the monitor sweep and recorder paper speeds by applying a signal of known frequency. Monitor sweep speed shall be one in/sec, plus or minus 10 percent and recorder paper speed shall be one in/sec, plus or minus 5 percent.

Internal calibration test. The one mV internal calibration pulse shall be accurate to plus or minus 15 percent.

CMMR test. Common mode rejection ratio shall be a minimum of 80 dB with respect to chassis ground with an imbalance of 65 kilohms between leads, and a 60 Hz impressed signal.

Overload test. Overload protection shall comply with ANSI/AAMI EC 13.

Frequency response test. Minimum monitor and recorder frequency response bandwidth shall be 1 Hz to 30 Hz, plus or minus 3 dB.

Charge time test. The time between activation of the charge control and full charge indication for the first and last of ten sequential charges and discharges, at maximum energy, shall not exceed 12 seconds, at 82.5°F.

Pulse energy test. The defibrillator shall be capable of delivering 10 pulses at maximum energy into a 50 ohm resistive load within 5 minutes at 32°F.

Synchromization time test. The time between the leading edge of a square pulse and of the defibrillator's synchronized discharge pulse shall not exceed 30 ms.

Battery capacity test. Battery capacity shall provide for a minimum of 2.25 consecutive hours of cardiac monitoring, or a minimum of 45 consecutive minutes of recording, or a linear combination of both. It shall also provide for a minimum of 30 shocks, at maximum energy, into a 50 ohm resistive load at 82.5°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Drip proof test. Place the unit in its normal operating position, with patient cable and defibrillator paddles attached, on a platform or tilt table in a manner such that it can be oriented horizontally and in any of four directions - forward, backward, and to each side - to the extent of 15 degrees from the horizontal plane.

Attach a spray nozzle with a uniformly distributed full cone spray pattern to a flexible hose and using a container calibrated in liters or an equivalent method, determine the flow rate. The flow rate shall be 16.3, plus or minus 1, liters/hour/sq. ft., three feet below the nozzle.

Position the nozzle three feet above the unit so that the spray is directed downward onto it. The unit shall be subjected to a uniform spray over the entire surface being tested. During the test, the item shall not be operated, but all controls shall be uncovered with the exception of the recorder.

Direct the spray onto the top of the unit (with platform or table horizontal) for at least eight minutes. Turn off spray for four minutes.

Repeat with the unit oriented 15 degrees from horizontal in the forward position. Repeat in the backward position. Repeat for both sides.

Return unit to horizontal and perform a full functional test.

Humidity test. Offerer shall supply the following documentation of compliance with the humidity test:

* Copy of test procedure
* Unit type
* Serial number
* Manufacturer and model number of chamber used
* Leakage current test results after 48 hours in the humidity chamber
* Energy accuracy test results after 48 hours in the humidity chamber

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Leakage current and energy accuracy test method.

Place the unit in a test chamber so that leakage current and energy accuracy tests can be performed external to the chamber.

Adjust the temperature of the chamber to 32°F.

Permit sufficient time for the temperature of the unit to stabilize.

Adjust relative humidity to 95 percent, plus or minus 5, and permit time to stabilize.

After 48 hours of conditioning, perform source, sink, and energy accuracy tests.

The unit is in compliance if leakages and accuracies are within specified limits.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

6515-01-204-1854

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

A-A-54894

MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA-MPQAS
USPHS
VA - OSS

Project Number: 6515-5495

Location: DEFIB17C.DOC

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-54894	23 July 1993

3. DOCUMENT TITLE
DEFIBRILLATOR, MONITOR-RECORDER (Battery Operated, with Charger)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial: AC 215-737-2870 (2) AUTOVON: AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:40Z
Check the source to verify that this is the current version before use.